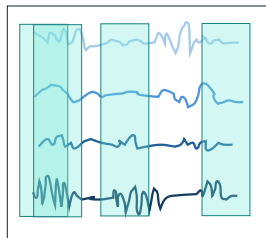


SIGReg: make every random slice look Gaussian

No teacher-student, no stop-gradient – just regularize the embeddings toward an isotropic Gaussian.

SIGReg regularization

Raw EEG Window



Local view extraction,
 $N_{local} = 4$.

LuMamba
Encoder

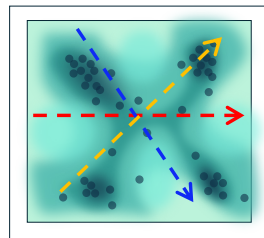


x_{local} v_{local}



● $v_{local,i}$ density deviation

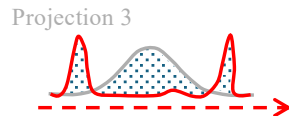
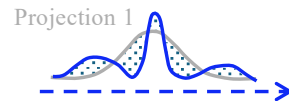
Embedding cloud



Projection onto $M = 300$
random directions.

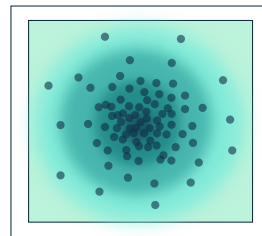
Over 2,000 samples per batch,
with N_{local} views per sample.

Epps-Pulley error between
each projected density and
a Gaussian.



LuMamba
Pre-training

Latent Space Evolution



If every 1-D projection is Gaussian, the whole cloud becomes an isotropic Gaussian → well-conditioned, collapse-free.